

SRETENSK, Russian Federation

WMO: 307770

Lat: 52.233N

Lon: 117.700E

Elev: 1723

StdP: 13.80

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-39.1	-35.0	-43.8	0.5	-38.8	-39.8	0.6	-34.7	21.6	0.6	15.9	1.6	2.6	240	0.631

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.0	87.4	64.3	83.5	63.6	79.9	62.6	68.2	79.0	66.5	77.0	65.0	75.1	6.0	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.9	98.3	72.7	63.2	92.6	70.5	61.6	87.3	69.0	33.7	79.0	32.3	76.5	31.0	75.2	74.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.3	15.3	13.0	-44.1	93.7	4.2	4.4	-47.1	96.9	-49.6	99.5	-51.9	102.0	-54.9	105.2		
(5)			-44.2	71.6	4.1	1.8	-47.2	72.9	-49.6	74.0	-51.9	75.0	-54.9	76.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.8	-16.5	-6.7	14.4	35.6	50.5	63.1	67.5	62.7	49.8	31.5	5.8	-14.3		
	DBStd	31.27	9.82	10.55	12.58	8.55	7.15	6.50	4.97	6.08	7.58	9.46	13.18	10.58		
	HDD50	9264	2060	1587	1105	439	81	2	0	2	95	574	1325	1994		
	HDD65	13418	2525	2007	1570	883	453	107	30	114	457	1038	1775	2459		
	CDD50	1528	0	0	0	6	95	396	544	397	89	1	0	0		
	CDD65	207	0	0	0	0	2	50	109	45	1	0	0	0		
	CDH74	2550	0	0	0	5	101	790	1128	490	36	0	0	0		
	CDH80	794	0	0	0	1	24	260	370	135	4	0	0	0		
	WSAvg	4.8	4.1	4.1	5.1	6.2	6.4	4.8	4.5	4.5	5.1	4.8	4.0	3.5		
	PrecAvg	13.9	0.2	0.2	0.2	0.6	1.1	2.1	3.7	3.4	1.5	0.4	0.3	0.2		
	PrecMax	20.7	0.4	0.6	0.6	1.7	3.8	4.8	8.4	6.3	4.5	1.2	0.9	0.7		
(15) Precipitation	PrecMin	8.0	0.0	0.0	0.0	0.1	0.2	0.2	1.3	0.7	0.1	0.0	0.0	0.0		
	PrecStd	3.0	0.1	0.1	0.2	0.4	0.9	1.2	1.6	1.3	1.0	0.3	0.2	0.2		
	DB	13.4	27.5	51.8	71.0	83.2	92.7	93.2	88.8	79.2	63.8	41.1	20.2			
	MCWB	10.7	22.6	38.6	47.9	56.9	66.1	65.6	65.8	59.0	47.1	33.2	17.7			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	6.5	21.0	43.9	63.0	76.2	86.5	88.0	83.7	72.5	55.7	33.8	11.2			
	2% MCWB	4.8	17.4	33.1	44.3	53.7	63.2	64.5	64.7	56.4	42.6	28.4	9.8			
	5% DB	2.4	15.5	38.4	56.9	71.8	82.2	83.9	79.4	68.2	50.2	29.0	5.4			
	5% MCWB	1.1	12.9	30.0	41.1	51.5	61.8	65.0	63.5	54.3	39.4	24.9	4.1			
	10% DB	-1.7	10.4	33.5	51.8	67.1	78.3	80.3	75.3	63.8	45.9	24.6	0.6			
	10% MCWB	-2.8	8.4	27.0	38.0	49.2	60.5	64.1	62.3	52.8	36.7	21.5	-0.3			
	0.4% WB	10.9	22.7	38.1	48.6	59.2	69.7	71.2	69.6	61.7	47.8	33.7	17.7			
	0.4% MCDB	13.2	27.3	50.6	69.2	78.2	85.0	82.9	80.9	70.6	63.1	40.6	19.9			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	4.8	17.3	33.5	44.8	55.9	66.4	69.0	67.2	59.1	43.5	29.0	9.8			
	2% MCDB	6.4	20.8	43.2	61.8	71.5	79.5	79.4	77.3	67.2	54.5	33.7	11.1			
	5% WB	1.2	12.9	30.4	41.6	53.0	64.2	67.3	65.5	56.9	40.1	25.0	4.3			
	5% MCDB	2.4	15.4	38.1	55.9	68.8	77.0	77.3	75.1	65.5	48.6	28.7	5.4			
	10% WB	-2.7	8.5	27.1	38.6	50.4	62.3	65.7	63.8	54.3	37.5	21.5	-0.2			
	10% MCDB	-1.8	10.4	33.4	50.8	65.0	74.1	75.5	72.6	62.4	45.1	24.5	0.6			
	MDBR	18.1	24.4	25.1	23.2	25.3	24.7	22.0	21.0	22.2	20.6	19.5	16.6			
	5% DB	19.2	25.6	26.6	33.8	35.0	33.5	30.4	29.1	30.2	27.0	21.9	18.4			
(36) Mean Daily Temperature Range	MCWBR	17.8	22.4	17.9	19.0	16.2	13.1	11.2	11.8	14.8	15.7	17.0	17.0			
	MCDBR	19.2	25.4	25.6	31.7	30.0	25.9	21.4	22.5	23.9	25.4	21.5	18.6			
	MCWBR	17.9	22.3	17.4	18.3	14.7	11.2	9.9	10.2	12.3	15.1	16.8	17.2			
	MCDBR	19.2	25.4	25.6	31.7	30.0	25.9	21.4	22.5	23.9	25.4	21.5	18.6			
(40) Clear-Sky Solar Irradiance	taub	0.247	0.272	0.317	0.420	0.497	0.462	0.417	0.390	0.350	0.311	0.275	0.253			
	taud	2.219	2.191	2.151	2.027	1.881	2.102	2.298	2.351	2.393	2.405	2.287	2.174			
	Ebn at Noon	242	268	275	256	241	252	263	265	263	252	227	213			
	Edh at Noon	21	29	37	49	59	48	39	35	30	23	19	18			
(44) All-Sky Solar Radiation	RadAvg	449	819	1333	1657	1806	1933	1757	1531	1192	807	470	339			
	RadStd	10	16	53	92	128	163	131	97	93	53	22	11			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page